

Advanced (Carolina Reaper)**Q1.** What is the expansion of $(x + y)^2$?

- (A) $x^2 + 2xy + y^2$
- (B) $x^2 - 2xy + y^2$
- (C) $x^2 + y^2$
- (D) $2xy$
- (E) $x^2 - y^2$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
- (B) $x^2 - 4x - 4$
- (C) $x^2 + 4x - 4$
- (D) $x^2 - 4x + 4$
- (E) $x^2 + x + 1$

Q3. What is the factored form of $x^2 + 6x + 9$?

- (A) $(x + 3)^2$
- (B) $(x - 3)^2$
- (C) $(x + 3)(x - 3)$
- (D) $(x + 2)(x + 4)$
- (E) $(x - 2)(x - 4)$

Q4. Which of the following is the expansion of $(x - y)^2$?

- (A) $x^2 + 2xy + y^2$
- (B) $x^2 - 2xy + y^2$
- (C) $x^2 - y^2$
- (D) $2xy$
- (E) $x^2 + y^2$

Q5. What is the factored form of $x^2 - 10x + 25$?

- (A) $(x + 5)^2$
- (B) $(x - 5)^2$
- (C) $(x + 5)(x - 5)$
- (D) $(x + 2)(x + 8)$
- (E) $(x - 2)(x - 8)$

Q6. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
- (B) $x^2 - 4x + 4$
- (C) $x^2 + 4x - 4$
- (D) $x^2 - 4x - 4$
- (E) $x^2 + x + 1$

Q7. What is the expansion of $(2x + 3y)^2$?

- (A) $4x^2 + 12xy + 9y^2$
- (B) $4x^2 - 12xy + 9y^2$
- (C) $4x^2 + 9y^2$
- (D) $12xy$
- (E) $4x^2 - 9y^2$

Q8. Which of the following is the factored form of $4x^2 + 12x + 9$?

- (A) $(2x + 3)^2$
- (B) $(2x - 3)^2$
- (C) $(2x + 3)(2x - 3)$
- (D) $(2x + 1)(2x + 9)$
- (E) $(2x - 1)(2x - 9)$

Open Ended Questions (FRQ)**Q9.** Factor the expression $x^2 + 8x + 16$ and show your work.**Q10.** Expand the expression $(x + 2y)^2$ and simplify.**Chef's Tips**

- Read each question carefully before answering.
- Use the space provided for free response questions.
- Check your work and show your calculations.